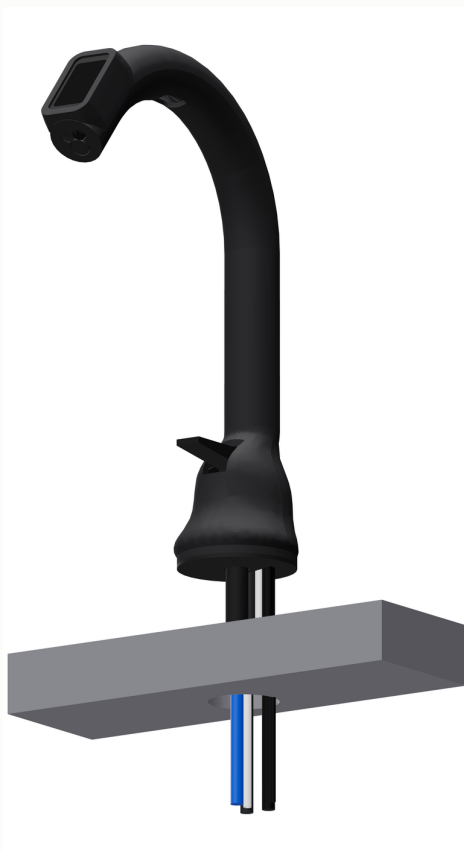


# Faucet assembly

An illustrated bench guide for the faucet, display and permanently attached umbilical.



## LOOKING FOR THE DISPLAY PINS?

## Start on page 2.

A labeled rear view, with USB-C at the top, shows the four pads and their main-board destinations. Page 3 shows the joint and the connection checks.

|                                            |    |
|--------------------------------------------|----|
| Rear display wiring                        | 2  |
| Connect and check the display              | 3  |
| Parts and tube cuts                        | 4  |
| Build the shell and vent chamber           | 5  |
| Fit display, mount hardware and supply     | 12 |
| Plug, sleeve, final inspection and packing | 15 |

Follow pages 4-17 in order for a complete build. Use pages 2-3 when you only need the wiring. The matching Sculpted or Industrial base, cover, plate and gasket surround the shared tip and display.

Coral identifies the part being added; blue identifies a tool, datum, wire path or check. Pictures are schematics unless identified as a CAD overview. Stated dimensions govern. The rear-board picture keeps the actual pad order.

### LETTER SHOP BOOK

18 pages, one-up, color, 100% / no scaling. The instructional layer is already centered at 98%. Source links and physical evidence scope are on page 18.

# The display wiring

Waveshare ESP32-S3-Touch-LCD-1.47. Hold the loose module with its back toward you and USB-C up.

**REAR / COMPONENT SIDE**  
Glass faces away from you

**+5 V to VBUS**  
P1 pin 1 · main board J3 V5

**0 V to GND**  
P1 pin 3 · main board J3 GND

**TX to main board RX**  
P1 pin 5 · GPIO43 to J3 IO35

**RX from main board TX**  
P1 pin 7 · GPIO44 from J3 IO33

**Top four LEFT pads**

USB-C UP ON BENCH

**INSTALLED: USB-C TOWARD THE DISPENSE FACE**

USB-C

dispense face

display

gooseneck

| DISPLAY PAD | FUNCTION           | MAIN BOARD J3 |
|-------------|--------------------|---------------|
| VBUS / P1-1 | +5 V power         | V5            |
| GND / P1-3  | 0 V return         | GND           |
| TXD / P1-5  | GPIO43, display TX | IO35, main RX |
| RXD / P1-7  | GPIO44, display RX | IO33, main TX |

Count P1 from the USB end: odd numbers left, even numbers right. This rear view is not mirrored.

**POWER AND SERIAL ARE DIFFERENT**  
VBUS takes 5 V. TX/RX are 3.3 V TTL at 921600 baud, 8N1. Leave VBAT, 3V3, RST and all other GPIO pads open. Disconnect J3 and USB while making the joints.

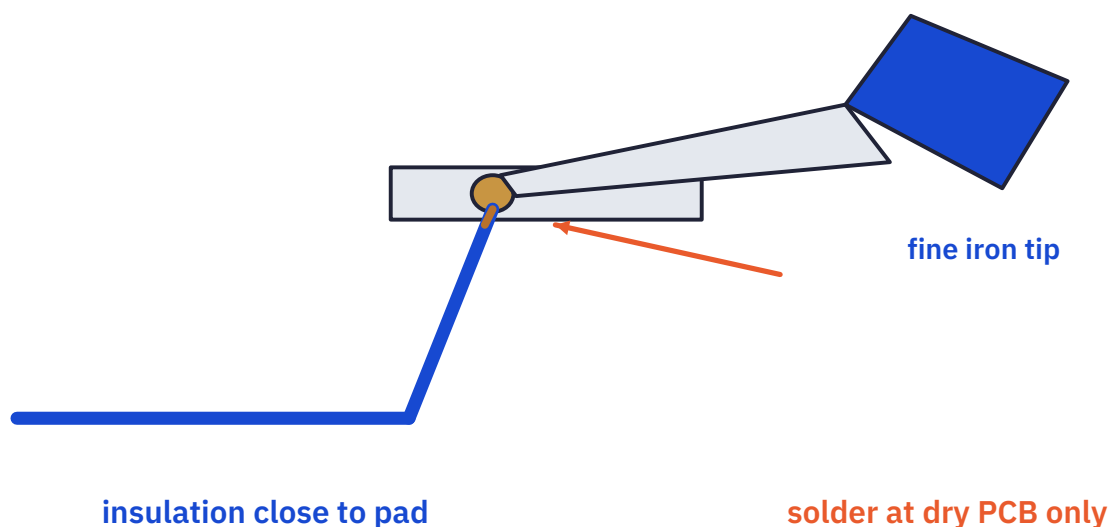
Waveshare P1 schematic + rear layout | faucet base\_link.cpp | appliance pins.h

LETTER / 2

# Connect and check the display

Use page 2 for the pad map. In a complete build, do this after the wires pass through both seated vent bungs (pages 8-10).

## AFTER BOTH BUNGS ARE SEATED



Detail section through one P1 pad. Make the electrical joints in the dry display pocket.

- 1 With J3 and USB disconnected, preserve the four lead identities. Strip only the dry PCB ends; solder each insulated lead directly to its page-2 P1 pad. Inspect for adjacent-pad bridges.
- 2 Leave free wire length for the page-12 slide and lowering motion. Route the leads in the open space below the PCB, clear of feet, components, lips and the USB socket.
- 3 With the harness disconnected at J3, trace V5 to VBUS, GND to GND, IO35 to TXD and IO33 to RXD through the actual plug/jack. Confirm there is no power-to-ground or adjacent-wire short.

## CHECK BEFORE CONTINUING

After the unpowered checks, power through J3. The screen should boot and receive the main board's selected flavor. A bright-screen tap changes the selection and the main board ticks; a dim-screen first tap only wakes it.

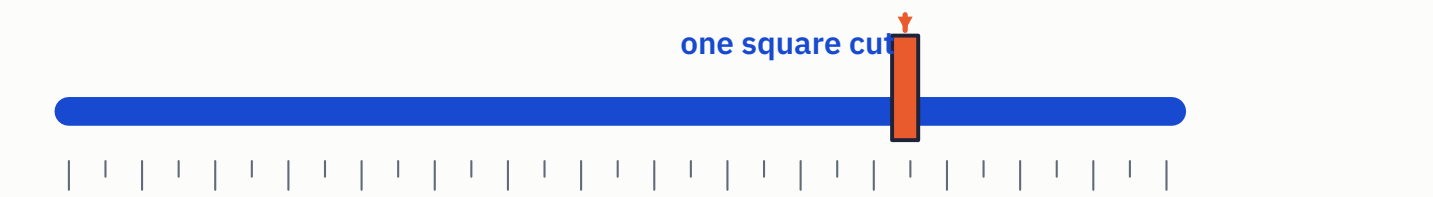
# Lay out the matching kit

Choose the style and finish before cutting. Keep the donor washer and nut loose, and both signal-cable ends unterminated.

**Rigid parts:** matching base, display cover and above-counter plate; shared shell tip; printed lever. **Soft parts:** matching countertop gasket, fresh TPU thimble and two vent bungs. **Hardware:** bare Westbrass, Waveshare 1.47 display, 3 short M3 inserts, 3 M3 × 8 screws, captive donor washer/nut, loose stainless under-counter plate.

## Factory tube cuts / mm

| TUBE                            | BLACK FAUCET | WHITE FAUCET |
|---------------------------------|--------------|--------------|
| Blue supply, 1/4-inch OD        | 1540         | 1540         |
| Soda tube, 3/8-inch OD          | 330          | 330          |
| White DRAIN, 4 mm OD            | 1872         | 1872         |
| Flavor A / B, black 1/4-inch OD | 1964 / 1964  | 1459 / 1500  |
| Flavor A / B, white 1/4-inch OD | -            | 496 / 455    |



### keep supplied reach

The 3/8-inch soda tube matches the faucet finish. Both flavor runs are black throughout a Black faucet. A White faucet has white upper runs and black lower runs joined by two PP0408W unions. The DRAIN is one continuous 1872 mm run.

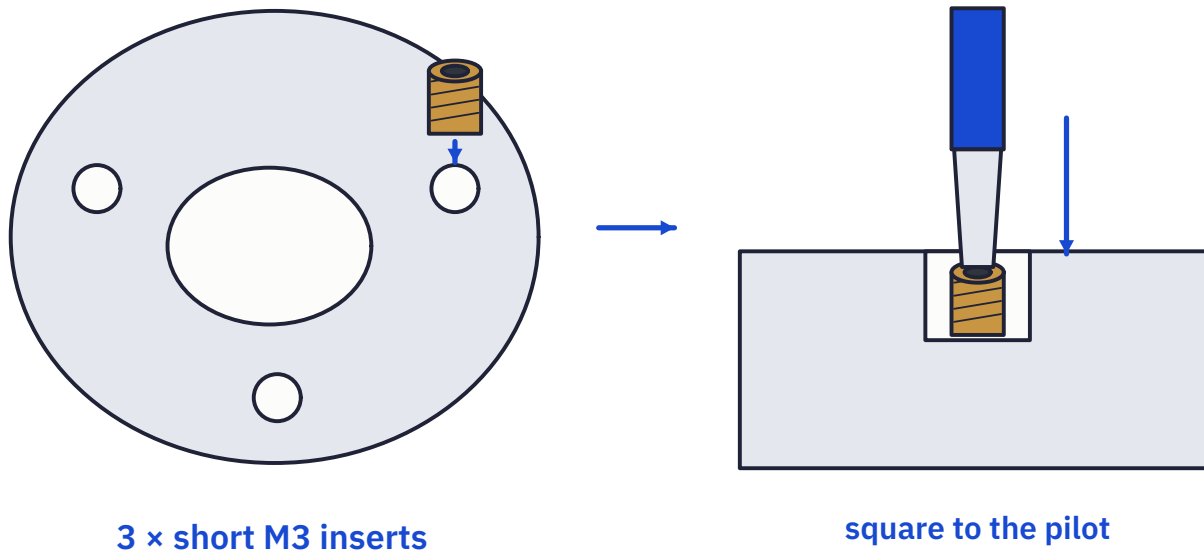
TOOLS AND CONSUMABLES

Tube cutter, 2.5 mm hex key, M3 heat-set tip/iron, fine soldering tip, solder, multimeter, magnifier, modular crimper and 3-prong 6P4C plug. Use the vent perimeter pusher, foam and PET braid already specified for this bench.

# Clean and fit the base inserts

Keep the shell base, tip and above-counter plate separate for access.

## SHELL FOOT / UNDERSIDE



Underside view at left; one insert/pilot section at right. The other two inserts use the same approach.

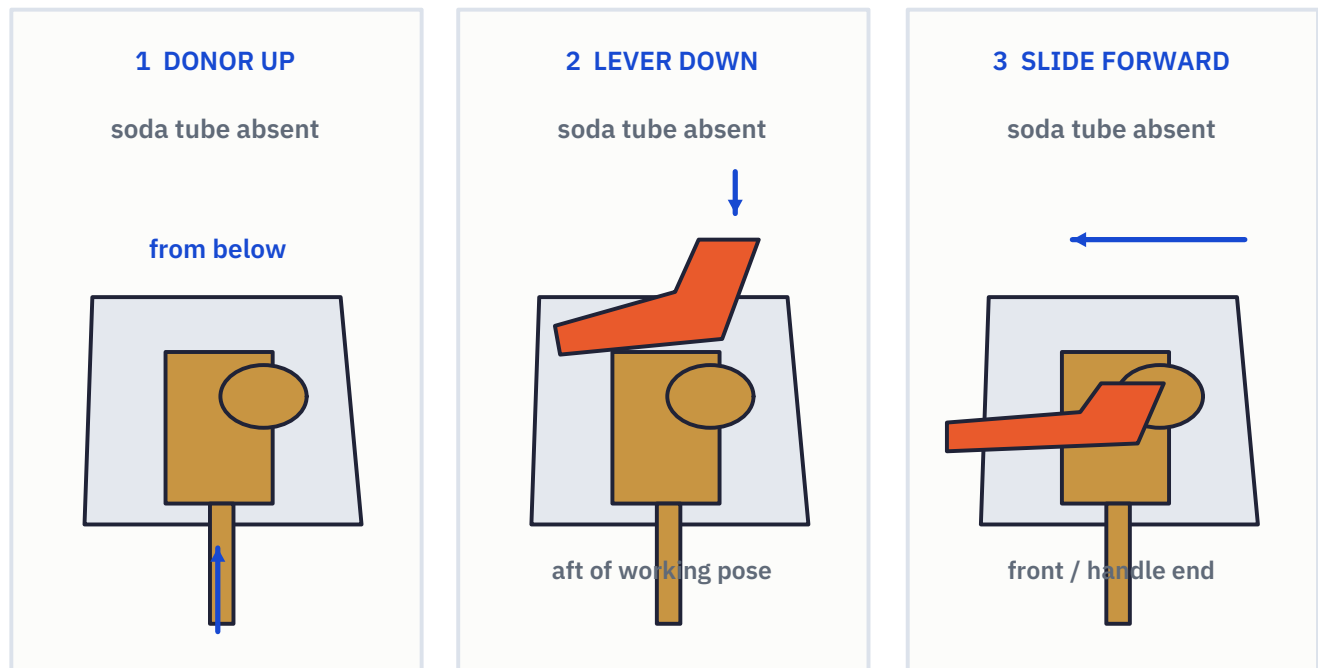
- 1 Remove supports and stringing from the donor cavity, tube passages, pedestal sockets and insert pilots. Dry-fit the matching printed pieces.
- 2 Heat-set the three ruthex RX-M3Sx4.0 short inserts into the base's bottom-facing  $\varnothing 4$  mm pilots. Keep each insert square; the specified insert-mouth datum is  $Z = 3.2$  mm.
- 3 Let the inserts cool undisturbed. Hand-start the M3 screws, then remove them; the plate is fitted after the tubes and shell joint.

### CHECK BEFORE CONTINUING

The pedestals enter their sockets freely, the mating surfaces close, and each cooled insert accepts its screw without forcing or turning in the print.

# Seat the donor, then the lever

The soda tube stays out during the complete aft/down/forward lever motion.



Side-view sequence. Front is the projecting handle end; the donor enters through the open foot.

- 1** Remove and retain the donor's own lever. With the base plate separate and soda tube absent, seat the bare donor from below, valve interface facing the front opening.
- 2** Hold the printed lever aft of its working position. Lower it onto the valve, then slide it forward around the metal cylinder until its light snap seats.
- 3** Move the handle through its actual travel. Check the rear arm's rising clearance before fitting the soda tube or closing the base.

## CHECK BEFORE CONTINUING

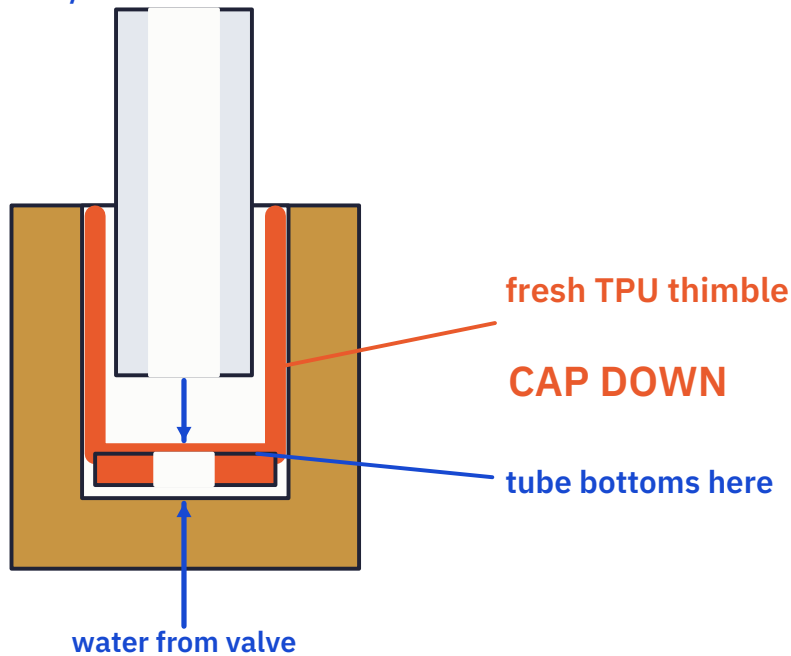
The lever seats and operates on the real donor without catching the shell. Keep the soda tube out if the lever needs to be removed or repositioned.

# Seat the thimble and soda tube

A fresh TPU thimble seals the 3/8-inch tube in the donor's top water port.

## SECTION THROUGH TOP PORT

3/8-inch soda tube



Section view. The annular cap has a center flow hole; the tube's square end seats on its upper face.

- 1 Place the fresh TPU 90A thimble cap-down in the donor's top port. The open end faces up toward the soda tube.
- 2 Feed the finish-matched 330 mm soda tube through the lower neck. Push its square-cut end down through the thimble until it bottoms on the cap.
- 3 Keep that seating while routing the tube toward the shared tip. The installed tube blocks the printed lever's aft disengagement path.

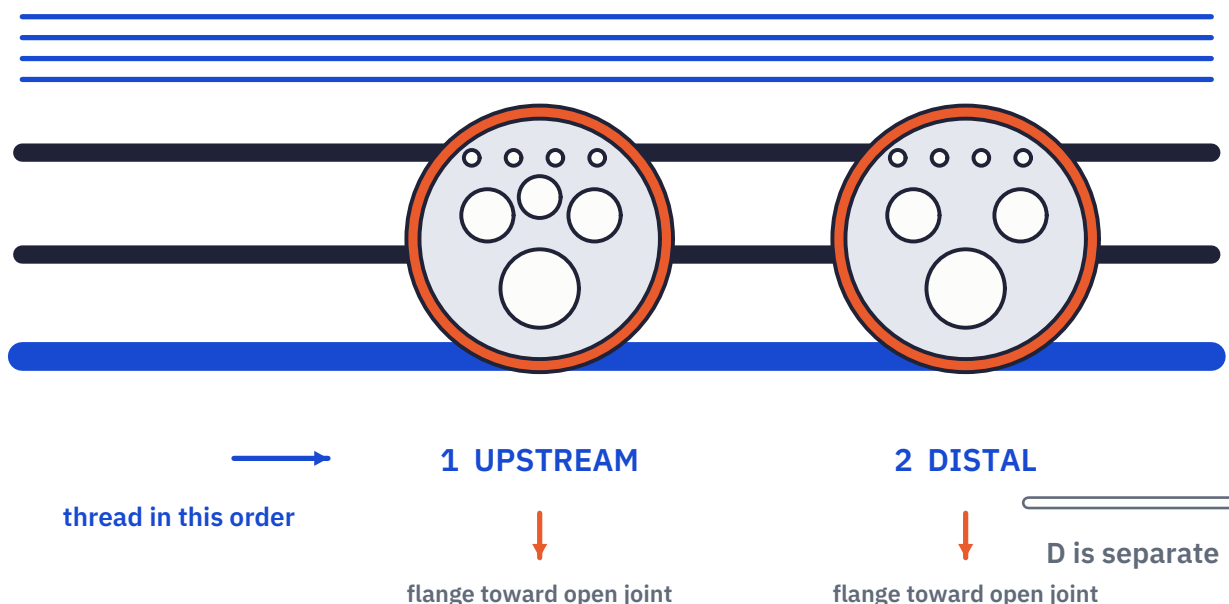
### CHECK BEFORE CONTINUING

The tube reaches its positive bottom seat, the lever still travels fully and the tube remains seated while the upper bundle is handled.

# Pre-thread the vent bungs

The display wires must be individually insulated through both seals before they are soldered.

## FREE DRINK / DISPLAY ENDS



Threading order, not an installed chamber section. Both bung flanges face back toward the open shell joint.

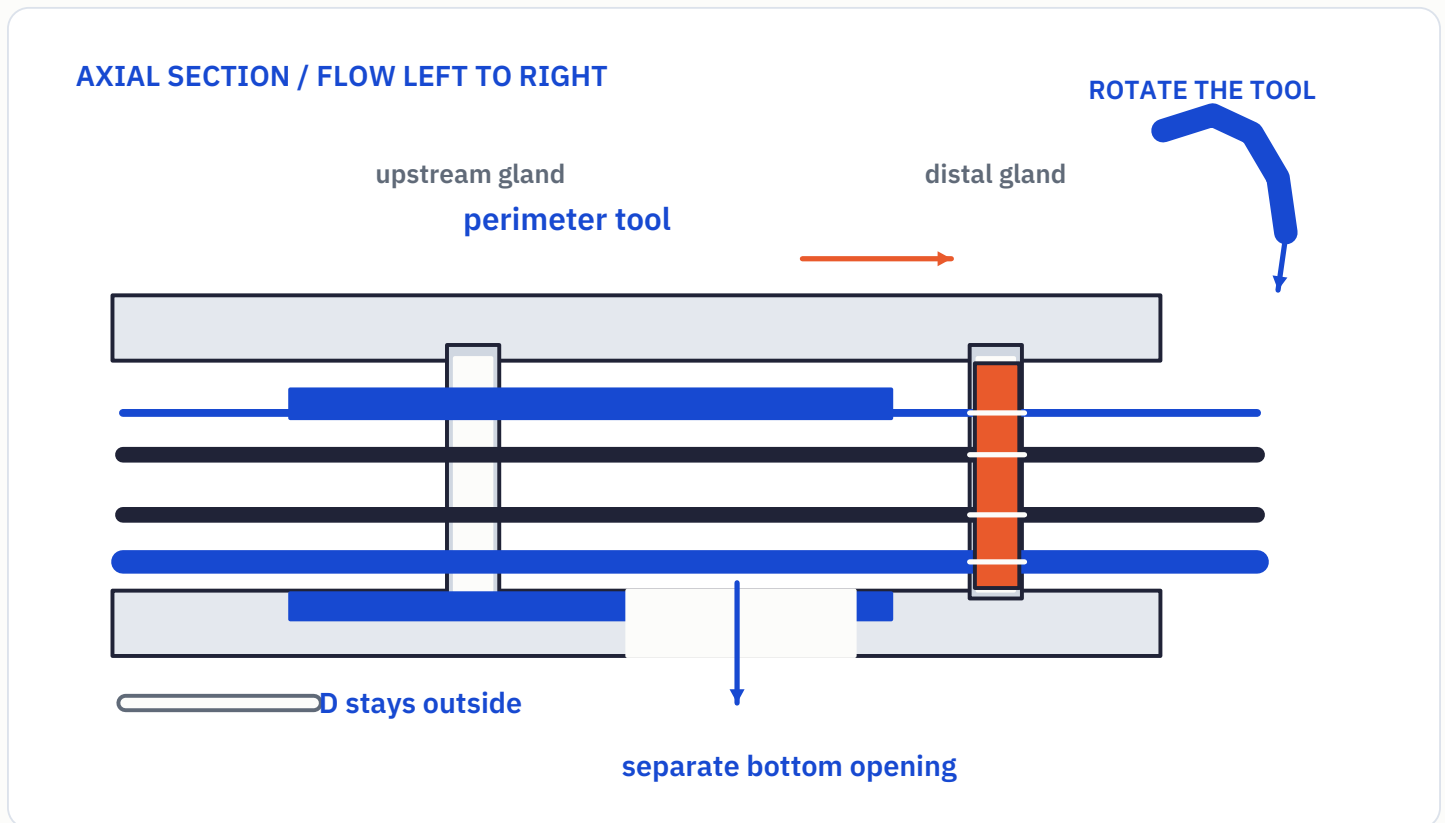
- 1 Keep the display ends unterminated. Peel the ribbon web into four continuously insulated conductors only through the local spread; preserve wire order and reject any nicked jacket.
- 2 Wet the bores with clean water. From the free S, F1, F2 and wire ends, thread the upstream bung first, then the distal bung. Leave the upstream D hole empty.
- 3 Park the upstream bung clear of the pusher's full stroke. Feed the three beverage ends and four wires through the tip's dry outlet guides, leaving D outside the tip.

## CHECK BEFORE CONTINUING

Four intact wire jackets and all three drink tubes pass through each bung. No stripped wire, solder joint, adhesive or slit is inside the seals.

# Seat the distal bung first

D stays outside while the slotted perimeter pusher passes through the empty upstream gland.



The large section is unwrapped for visibility. The inset shows the actual curved insertion direction.

- 1 Fit the pusher around the continuous bundle through its 19 mm side slot. Its leading annulus bears on the distal bung's perimeter.
- 2 Rotate the curved tool about the gooseneck arc center. Advance the distal bung through the empty upstream gland until its body meets the distal backstop and its flange enters the retaining groove.
- 3 Withdraw the tool along that same curve until it clears the socket, then lift it off through the open side. Leave the upstream bung parked outside.

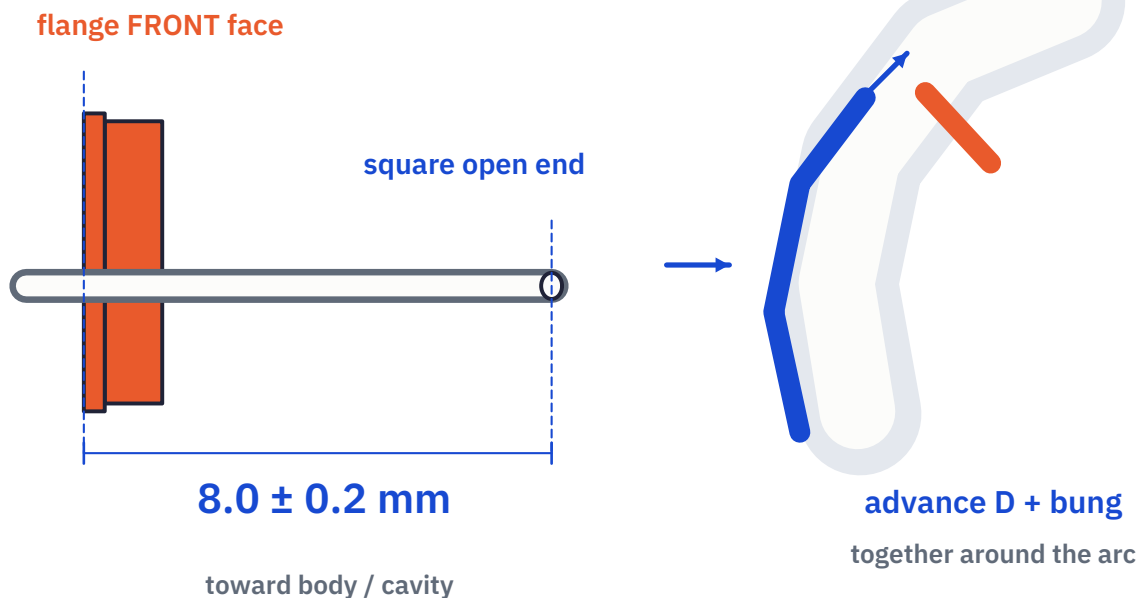
## CHECK BEFORE CONTINUING

The complete distal flange is captured, its body is seated at the backstop, and the continuous tubes and insulated wires remain in their assigned bores.

# Set D and seat the upstream bung

The drain ends inside the wet chamber. It does not continue to the drink face.

## SET D WHILE THE BUNG IS OUT



Measure from the flange front face contacted by the tool, toward the bung body and cavity, to D's square cut.

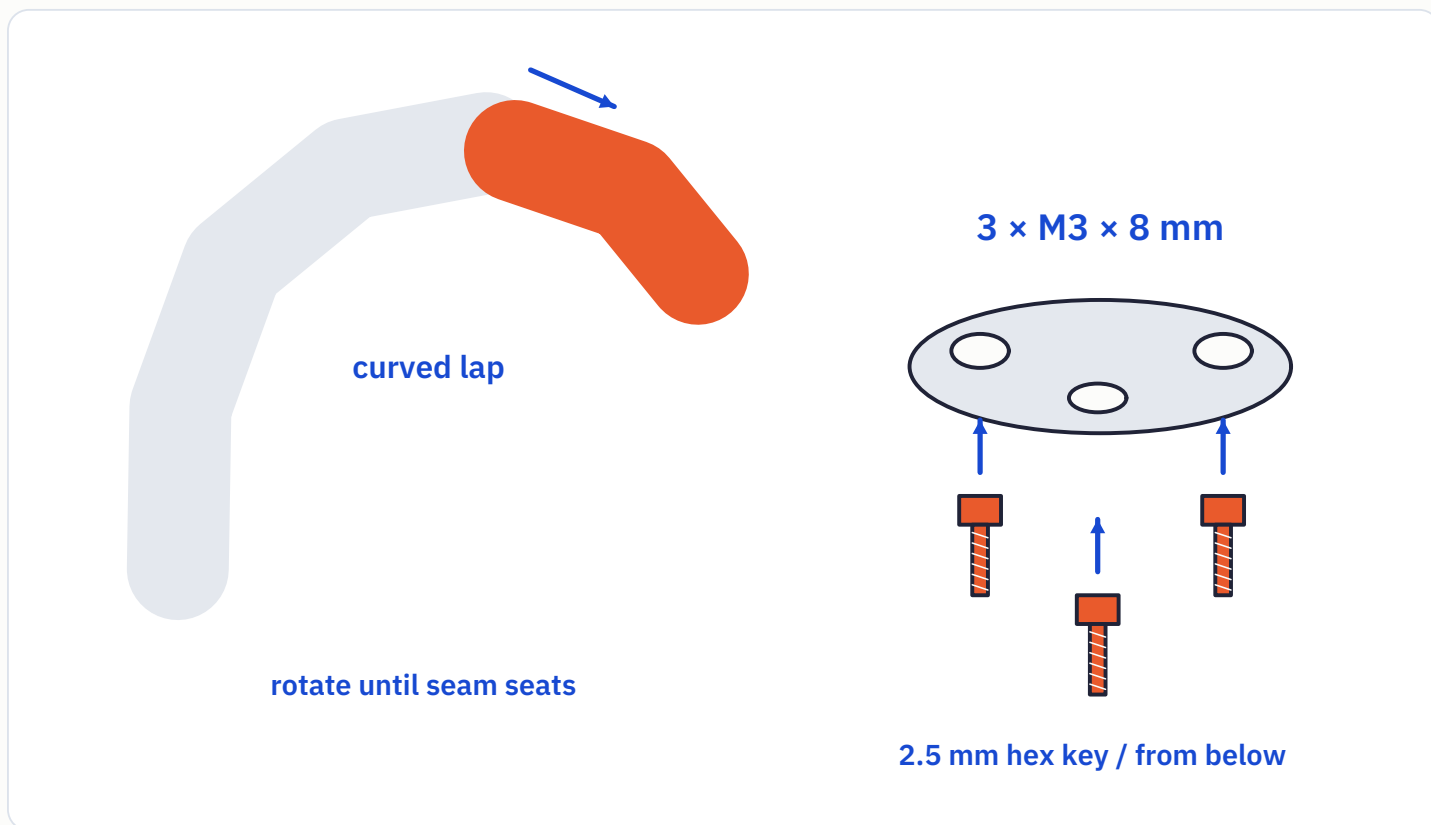
- 1 With the upstream bung outside the tip, feed D through its own hole. Set the square-cut end to  $8.0 \pm 0.2 \text{ mm}$  from the flange's front face toward the body/cavity.
- 2 Refit the perimeter pusher and rotate D plus the upstream bung into the gland together. Its flange seats in the groove against the body-seat shoulder; the upstream body has no terminal backstop.
- 3 Remove the pusher by the same curved withdrawal and side exit. Confirm D opens into the cavity and the separate bottom discharge opening is completely clear.

## CHECK BEFORE CONTINUING

Both flanges are captured; every intended tube/wire bore is occupied. D's open end projects into the cavity, while the three drink tubes continue through the distal seal into their dry channels.

# Close the neck and base

Route the continuous bundle before rotating the dry curved lap closed.



Neck joining at left; base screws aligned with their underside holes at right. Schematics, not screw-station dimensions.

- 1 Rotate the tip socket over the base plug about the arc center until the seam seats. Preserve the soda tube's donor-port seat and keep the chamber's bottom opening clear.
- 2 Pass F1/D/F2 and the flat ribbon through the plate's rear passage, and the bare shank through its center. Keep the ribbon flat behind the bundle. Seat all three pedestals.
- 3 Install three M3 × 8 screws from below with a 2.5 mm hex key, progressively and evenly. Square-trim only the three beverage outlets flush with their symmetric drink face.

## CHECK BEFORE CONTINUING

The plate and curved seam close evenly. Lever travel and both flavor paths remain free. D ends inside the chamber and its separate bottom opening remains bare. Now connect/check the display using pages 2-3.

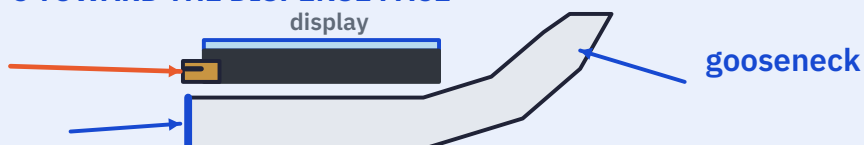
# Fit the display and cover

USB-C faces the dispense face; the opposite end points up the gooseneck. Complete the page-3 joints/check first.

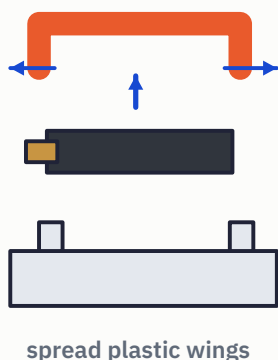
## INSTALLED: USB-C TOWARD THE DISPENSE FACE

USB-C

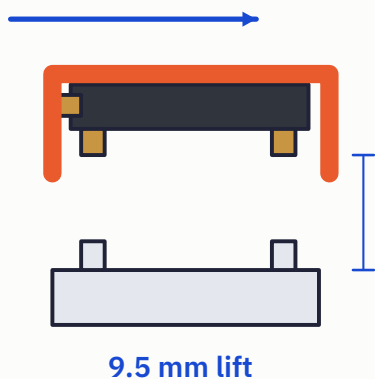
dispense face



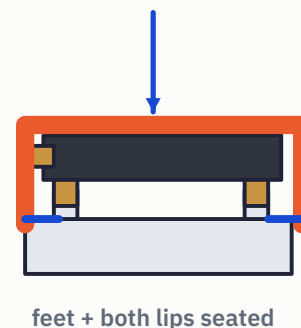
### 1 DISPLAY INTO COVER



### 2 SLIDE FROM OUTLET



### 3 LOWER TO SEAT



Side sections show the USB-C end at the outlet. Two of four feet are visible; the 9.5 mm lift is normal to the glass.

- 1 Keep USB-C toward the dispense face. Route the leads freely below the PCB toward its southwest corner as viewed from the glass. Spread the wings and load the display through the cover's open underside.
- 2 Hold the pair 9.5 mm above the final seat, normal to the glass. Slide it along the tip from the outlet end until the four metal feet align with their supports; feed the ribbon as it moves.
- 3 Lower the pair normal to the glass until both broad lips seat in the side grooves. Check all four feet, the glass/bezel clearance and the real tube/wire/component clearances.

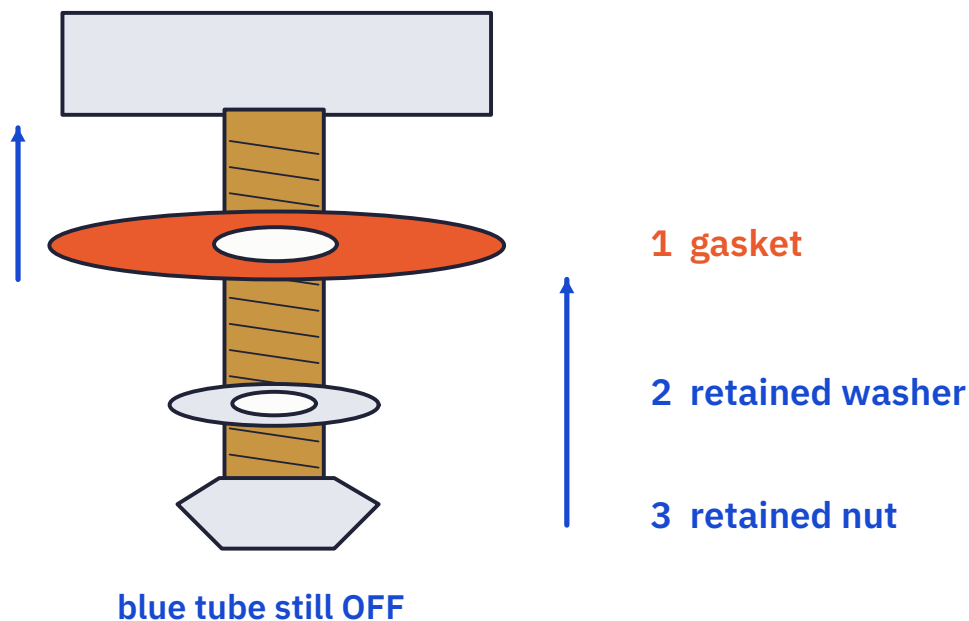
## CHECK BEFORE CONTINUING

Both retaining lips and four feet are seated, the seam closes and touch pressure leaves the display seated. No wire is pinched or pulling a solder pad. The display has no mounting screw or insert.

# Fit gasket, washer and captive nut

The mount hardware must be on the bare shank before the blue supply is permanently connected.

## FACTORY ORDER / SHANK BELOW



Exploded order from the shell foot downward. The steel under-counter plate stays loose for the install bag.

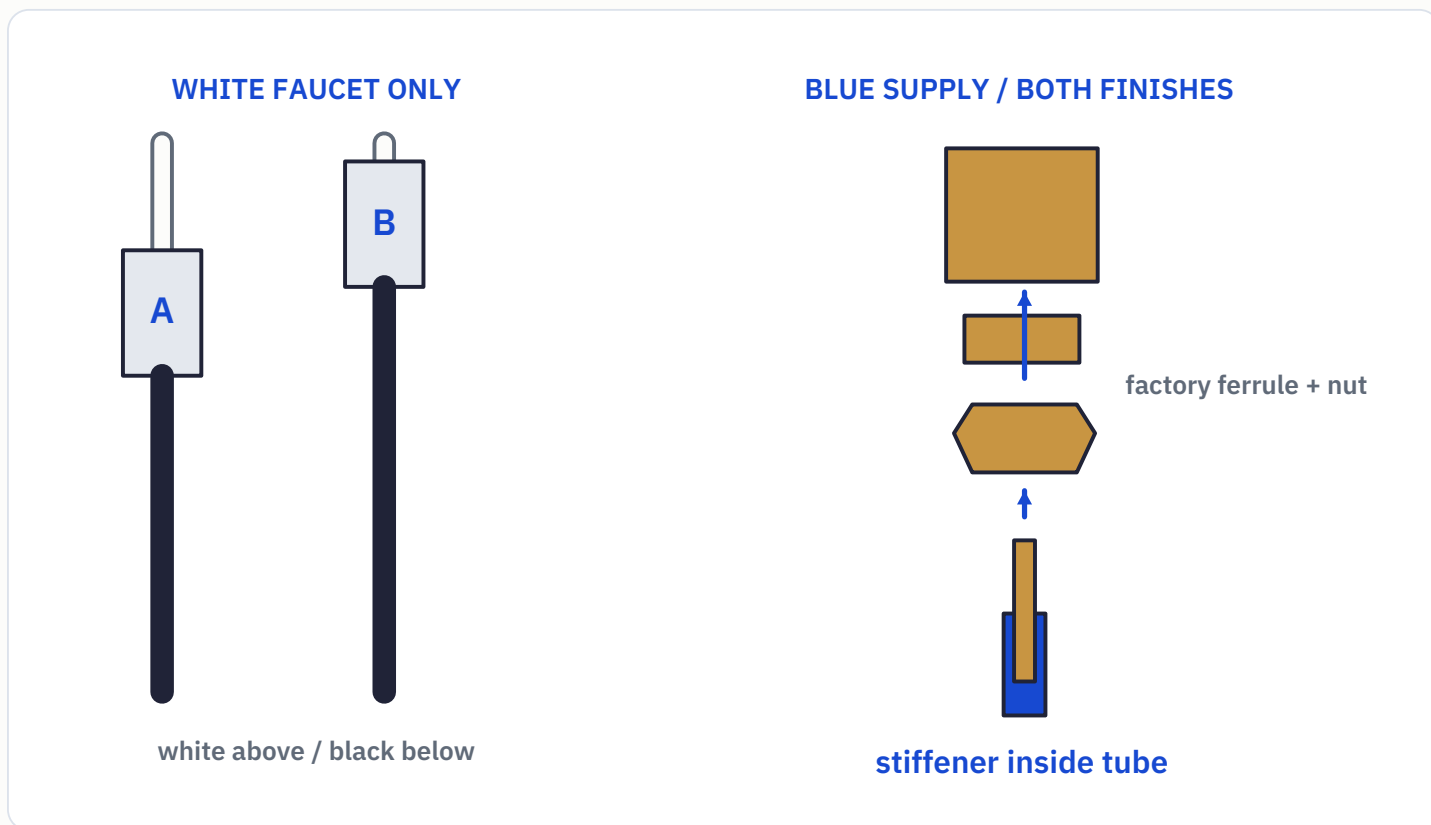
- 1 Pass both flavor tails, D and the unterminated wall end of the ribbon through the gasket's matching passage. Pass the bare shank through its center and slide the gasket flat against the printed plate.
- 2 Slide the retained donor washer onto the shank, then thread on the retained donor nut loosely. Keep the gap above them for the countertop and laterally inserted steel plate.
- 3 Leave the under-counter steel plate off. The next page connects the blue tube below the hardware, making this washer and nut permanently captive.

## CHECK BEFORE CONTINUING

The gasket lies flat and covers the base screw heads. The washer/nut are on the shank above the supply joint. The 38 mm routing envelope does not establish maximum clampable countertop thickness.

# Join White runs and the blue supply

A Black faucet has continuous flavor tubes. Both finishes use the same donor supply connection.



White flavor unions at left, B above A. Blue supply connection at right, shown before tightening.

- 1 On a White faucet, push each white upper flavor run and matching black lower run into its PP0408W union to the 16 mm stops. Keep B's union one 30 mm union length above A's, at the factory stations.
- 2 Keep F1/D/F2 and ribbon straight through the countertop routing zone. Below that zone, preserve the specified R30 flavor/ribbon returns and R25 drain returns around the unions.
- 3 Put one Siptenk brass stiffener fully inside the blue 1/4-inch tube end. Insert the stiffened end into the donor's lower compression port and tighten its factory ferrule/nut hand-snug plus 1/4 turn.

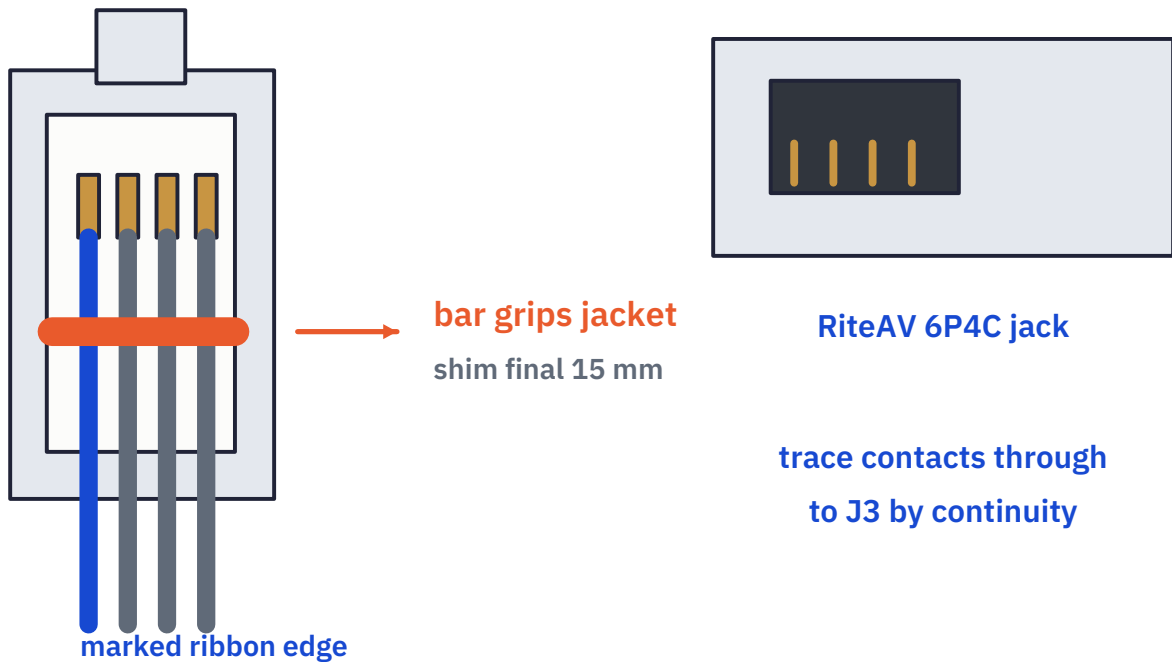
## CHECK BEFORE CONTINUING

Both White-faucet unions bottom and pass a tug check. The blue tube is connected below the captive hardware. The separate 3/8-inch soda tube remains in the donor's top port.

# Finish the wall plug by continuity

The four nets are on page 2. Match the actual modular plug and jack to those endpoints.

## CRIMP ON THE WALL END



Four center contacts and jacket strain-relief bar. The picture does not assign a telephone color code or modular-contact net order.

- 1 Crimp a 3-prong RJ11 6P4C plug on the wall end, conductor 1 on the ribbon's marked edge. The strain-relief bar must grip the jacket; shim its final 15 mm with tape or heat-shrink as specified.
- 2 With power removed and J3 unplugged, identify which actual jack contact each display lead reaches. Terminate the inboard J3 loom so V5, GND, IO35 and IO33 reach VBUS, GND, TXD and RXD respectively.
- 3 Record the four contact-to-net identities for the build. Trace through the mated plug/jack and check adjacent conductors. Use contact labels and continuity, including any reversal across the connector.

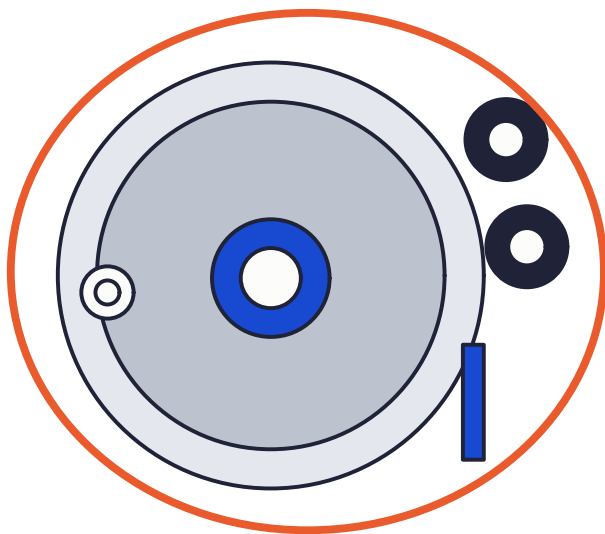
## CHECK BEFORE CONTINUING

The page-2 four-net map holds end to end through the mated connector, and the crimp bar carries cable tension. A telephone/T568B color diagram does not define SIG-6.

# Insulate and sleeve the bundle

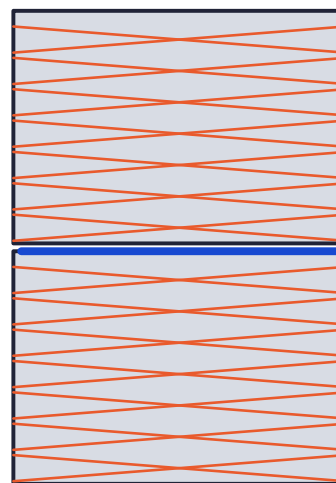
Lay the completed signal cable beside all four tubes before sliding on any braid segment.

FOAM ON BLUE ONLY



common PET braid

D opposite the flavors



butt adjoining segments

fit braid before cutting

Bundle section at left; two fitted sleeve segments at right. D gathers opposite the cold foam from the flavors.

- 1 Fit foam only to the blue cold tube, one segment at a time, butting the segment above it. The standard run uses five pieces covering 1322 mm; leave 143 mm bare at the top and 75 mm at the wall tail.
- 2 Gather the flavor pair beside the foam and D on its opposite side, with SIG-6 inside the braid. Slide one fitted PET-braid segment over the pack for each foam segment.
- 3 Fit before cutting: braid shortens when expanded. The top braid extends 126 mm above the foam to cover both unions on a White faucet, or that same length of the continuous Black-faucet runs.

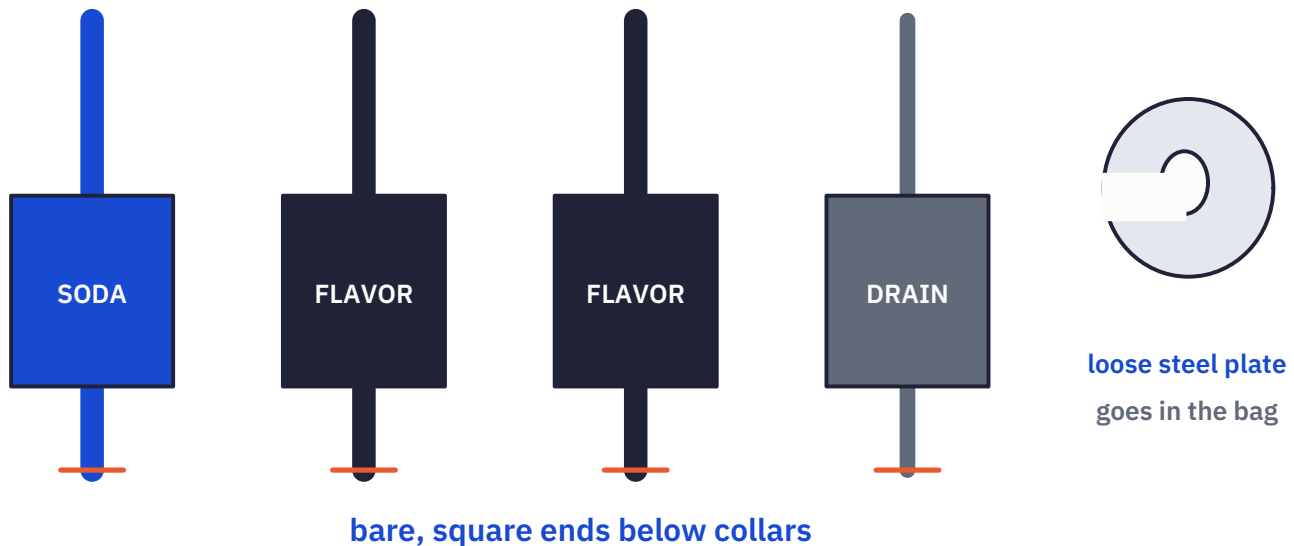
## CHECK BEFORE CONTINUING

Foam joints and braid joints butt without gaps; no braid segment excludes the display cable or D. All four tails remain bare, square and accessible beyond the final sleeve.

# Identify, inspect and bag

The customer receives the completed faucet, gasket, captive mount hardware, tubes and signal plug.

## IDENTIFIED WALL TAILS



Four identified bare wall tails. One stainless under-counter plate ships loose beside the complete faucet/umbilical.

- 1 Thread the matching collars onto the wall tails: blue SODA, two black FLAVOR and white 4 mm DRAIN. Run each collar up the bare tail below the braid with its word visible.
- 2 Inspect the complete unit: seams/lips seated; full lever motion; three beverage ends flush; D and bottom port clear; unpinched cable; page-2 continuity and page-3 powered flavor response.
- 3 Coil the umbilical to an 8-12 inch loop diameter, bag it with the loose steel plate, then seal and label with the build number and FAUCET-UMBILICAL-SUBASSEMBLY.

## CHECK BEFORE CONTINUING

The blue supply is permanently attached below the captive washer/nut. The gasket is already fitted. The loose steel plate is in the bag. Acceptance of vent containment and mounting range follows the linked physical qualification, not this visual inspection.

# Sources and fit scope

The guide adds the order and pictures. These procedures, manufacturer documents and physical records own the requirements.

**Faucet and umbilical: cuts, mount stack, braid and packing**

OPEN SOURCE

**Shell assembly: donor, lever, thimble and display seating**

OPEN SOURCE

**Vent seals: threading, curved pusher and drain setting**

OPEN SOURCE

**Waveshare: actual rear layout and interface labels**

OPEN SOURCE

**Waveshare: P1 numbers and power/UART schematic**

OPEN SOURCE

**Faucet firmware: GPIO43 TX, GPIO44 RX, 921600 baud**

OPEN SOURCE

**Main board: J3 IO33 TX and IO35 RX**

OPEN SOURCE

**SIG-6: the inboard loom and rear jack**

OPEN SOURCE

## Physical evidence has a specific scope

The recorded display cover has accepted PET-GF give, spring, fit and retention; the recorded lever has accepted fit and operation. These results belong to their identified physical articles. Retention force, cycle life, the current complete vent assembly's containment and maximum clampable countertop thickness remain separate qualifications.

**Display cover acceptance**

**Lever acceptance**

**Vent-cavity qualification**

Manufacturer layout and schematic reviewed 2026-10-08. The source-hash receipt is committed beside the delivery copy. Rebuild by hand when deliberately reviewing this guide; no machine build regenerates it.